Generative Adversarial Networks (GAN)

Generative models: outputs – objects from real world: images, texts, audios, ...

Dataset: $\{x_1, x_2, \ldots, x_N\}$ – images from MNIST

Goal: $x \sim p(x)$

GAN: $z \in \mathbb{R}^d$, $z \sim \mathcal{N}(z|0, I)$

$x = g(z, \theta)$ – generator

e.g. conv decoder

$p(x|\theta)$

Typical loss in gen. models: $-\frac{1}{N}\sum_{i=1}^{N} \log p(x_i|\theta) \to \min_\theta$

$p(x|\theta) = \int \delta(x - g(z,\theta))\, p(z)\, dz$ – can't be computed!

(δ ← delta-function; $p(z) = \mathcal{N}(z|0, I)$)

$$\delta(x-a) = \begin{cases} +\infty, & \text{if } x = a \\ 0, & \text{if } x \neq a \end{cases}$$

$$\int_{-\infty}^{+\infty} \delta(x-a)\,dx = 1 \qquad \delta[f] = \int f(x)\,\delta(x-a)\,dx = f(a)$$

discriminator $d(x) \rightarrow$ "real" / "fake" $\quad d(x) \in (0, 1)$

Fixed g: $\mathbb{E}_{x \sim p_{data}(x)} \log(d(x)) + \mathbb{E}_{z \sim N(z|0,I)} \log(1 - d(g(z)))$ $\rightarrow \max_d$

Fixed d: $\underbrace{\mathbb{E}_{x \sim p_{data}(x)} \log(d(x))}_{\text{const}} + \mathbb{E}_{z \sim N(z|0,I)} \log(1 - d(g(z)))$ $\rightarrow \min_g$

GAN scheme

$d(x, \theta_d)$, $g(z, \theta_g)$

Init θ_d, θ_g

Until convergence:

maybe several times

$$\begin{bmatrix} x_1, \dots, x_M \sim P_{data}(x) \\ z_1, \dots, z_M \sim N(0, I) \\ L_d = -\frac{1}{M} \sum_{i=1}^{M} \left(\log d(x_i, \theta_d) - \log\left(1 - d(g(z_i, \theta_g), \theta_d)\right)\right) \\ \theta_d = \text{opt_step}(\nabla_{\theta_d} L_d) \end{bmatrix}$$

$$\begin{bmatrix} \tilde{z}_1, \dots, \tilde{z}_M \sim N(z \mid 0, I) \\ L_g = \frac{1}{M} \sum_{i=1}^{M} \log\left(1 - d(g(\tilde{z}_i, \theta_g), \theta_d)\right) \\ \theta_g = \text{opt_step}(\nabla_{\theta_g} L_g) \end{bmatrix}$$

$$V(d,g) = \int p(x) \log(d(x))\,dx + \int q(x) \log(1-d(x))\,dx$$

(data distr. $p(x)$; gen. distr. $q(x)$)

Fix g: $V(d,g) \to \max_d$

Fix x: $p \log(d) + q \log(1-d) \to \max_{d \in (0,1)}$

$$\frac{\partial}{\partial d} = \frac{p}{d} - \frac{q}{1-d} = 0 \Rightarrow p - pd - qd = 0$$

$$\Rightarrow d = \frac{p}{p+q}$$

Opt. $d(x) = \frac{p(x)}{p(x)+q(x)}$

$$V\left(d(x)=\frac{p(x)}{p(x)+q(x)},\ g\right)=\int p(x)\log\frac{p(x)}{\frac{1}{2}(p(x)+q(x))}dx+$$

$$+\int q(x)\log\left(1-\frac{p(x)}{p(x)+q(x)}\right)dx+2\log\frac{1}{2}=$$

$JS(p_{data}\,\|\,g)\to\min_g$
$\Rightarrow g(x)=p_{data}(x)$

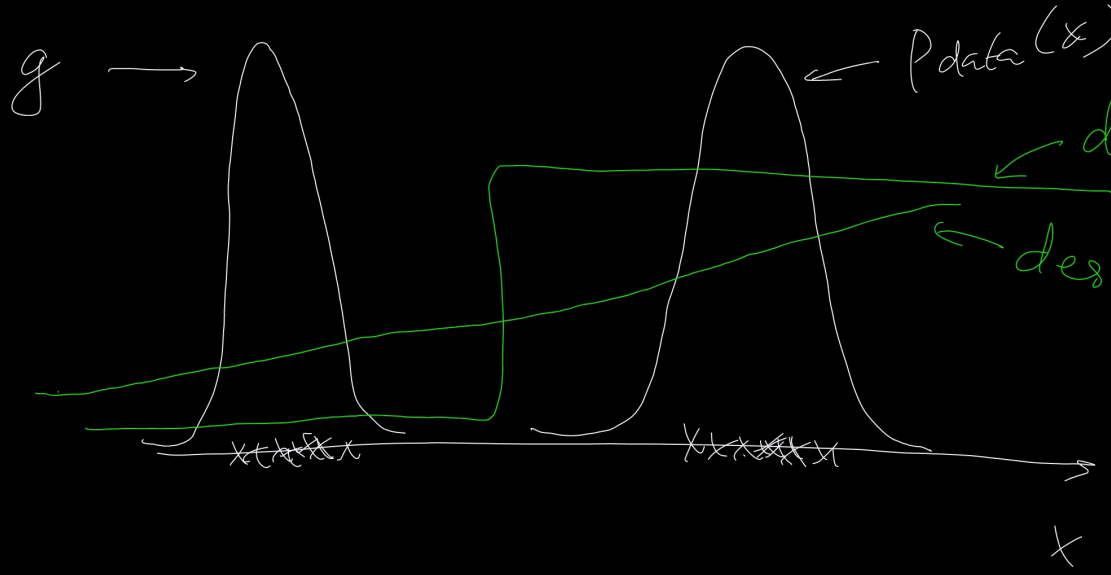
$$\frac{q(x)}{\frac{1}{2}(p(x)+q(x))}$$

Jensen-Shannon divergence

$$=KL\left(p\,\|\,\frac{1}{2}(p+q)\right)+KL\left(q\,\|\,\frac{p+q}{2}\right)=2JS(p\|q)$$

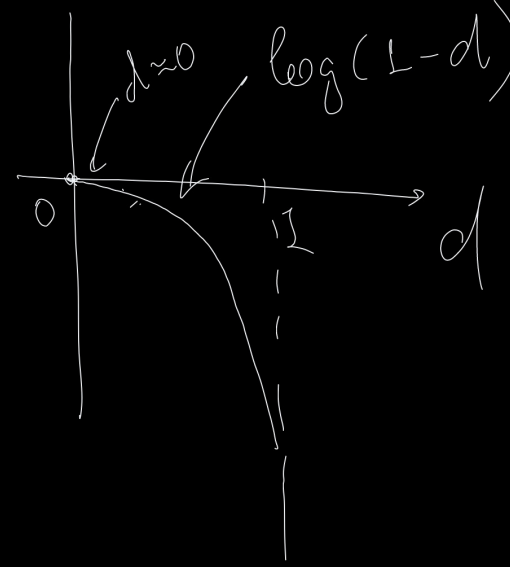
$JS(p\|q)\geq 0 = 0\Leftrightarrow p=q$

Training classic GAN is unstable!

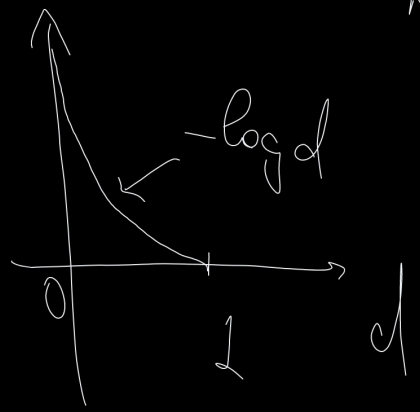


Trick 1: use unsaturated loss for generator

$$\mathbb{E}_z \log(1 - d(g(z))) \to \min_g$$



Unsaturated loss: $-\mathbb{E}_z \log d(g(z)) \to \min_g$



Trick 2: Label Smoothing for discr.

"real" – 1 → 0.9 "real"

→ 0.1 "fake"

$$L_d = -0.9\,\mathbb{E}_{x\sim p_{data}(x)} \log d(x) - 0.1\,\mathbb{E}_{x\sim p_{data}} \log(1-d(x)) - \mathbb{E}_z \log(1-d(g(z))) \to \min_d$$

Trick 3: Noise injection

$$\varepsilon, \tilde{\varepsilon} \sim N(0, \sigma^2 I)$$

$$L_d = -\mathbb{E}_{x \sim p_{data}(x)} \log d(x + \varepsilon) - \mathbb{E}_z \log\left(1 - d(g(z) + \tilde{\varepsilon})\right) \to \min_d$$

Modern GANs

1. Loss: Least Squares, Hinge loss, Wasserstein
2. Discriminator: Spectral Normalization, Adaptive Discr. Augmentation (ADA)
3. Generator: Noise injection on diff. resolution levels, AdaIN, Mapping Network

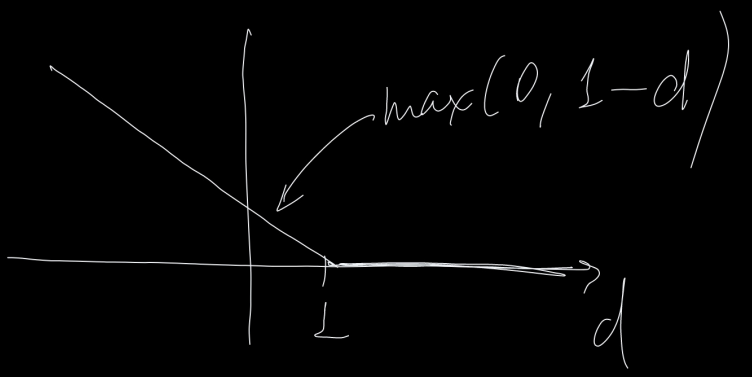
Least Squares GAN

$d(x) \in \mathbb{R}$ $(\notin (0,1))$

$$L_d = \mathbb{E}_{x \sim p_{data}(x)} \left(d(x) - 1\right)^2 + \mathbb{E}_z \left(d(g(z)) - 0\right)^2 \to \min_d$$

$$L_g = \mathbb{E}_z \left(d(g(z)) - 1\right)^2 \to \min_g$$

Hinge Loss GAN

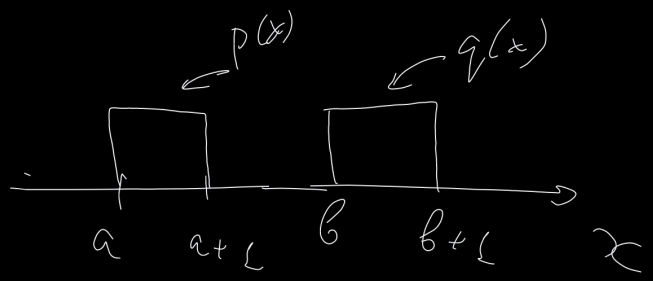


$$L_d = \mathbb{E}_{x \sim p_{data}} \max(0, 1 - d(x)) +$$

$$+ \mathbb{E}_z \max(0, 1 + d(g(z))) \to \min_d$$

$$L_g = -\mathbb{E}_z \, d(g(z)) \to \min_g \qquad d(g(z)) \le -1$$

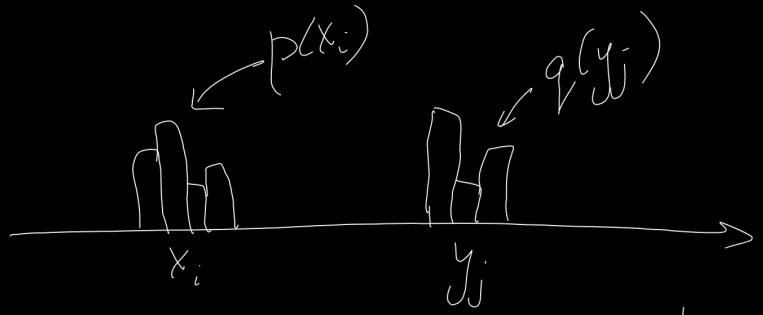
Wasserstein distance



$$W(p,q) = |b-a|$$

$$KL(p\|q) = \int p(x) \log \frac{p(x)}{q(x)} dx = \infty$$

$$JS(p\|q) = \frac{1}{2}\left(KL\left(p \,\|\, \frac{p+q}{2}\right) + KL\left(q \,\|\, \frac{p+q}{2}\right)\right) = \frac{1}{2}\int_a^{a+1} 1 \log \frac{1}{1/2} dx + \frac{1}{2}\int_b^{b+1} 1 \cdot \log \frac{1}{1/2} dx = -\log \frac{1}{2}$$



$\gamma(x_i, y_j)$ – transportation plan

$\|x_i - y_j\|$ – transportation cost

Linear Programming →

$$\begin{cases} \sum\limits_{x_i, y_j} \gamma(x_i, y_j)\|x_i - y_j\| \to \min\limits_{\gamma} \\ \gamma(x_i, y_j) \geq 0 \quad \forall i, j \\ \sum\limits_{x_i} \gamma(x_i, y_j) = q(y_j) \\ \sum\limits_{y_j} \gamma(x_i, y_j) = p(x_i) \end{cases}$$

$$W(p,q) = \inf_{\gamma \in \Pi(p,q)} \mathbb{E}_{x,y\sim\gamma} \|x-y\|_1$$

coupling

$$\gamma \in \Pi(p,q) \Leftrightarrow \gamma(x,y) \geq 0 \quad \forall x,y$$

$$\int \gamma(x,y)\,dx\,dy = 1$$

$$\int \gamma(x,y)\,dx = q(y)$$

$$\int \gamma(x,y)\,dy = p(x)$$

Th (Kantorovich–Rubistein duality)

$$W(p,q) = \max_{f \in L\text{-Lipschitz}} \left(\mathbb{E}_{x \sim p} f(x) - \mathbb{E}_{x \sim q} f(x) \right)$$

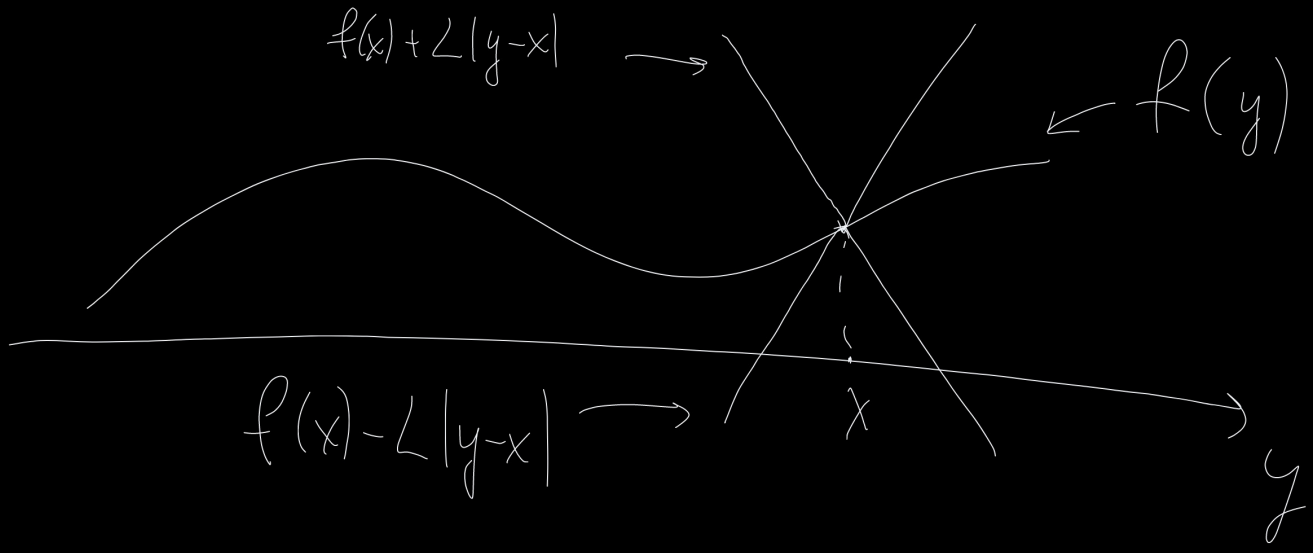
$|f'(x)| \le L \quad \forall x$

$\Downarrow$

$f \in L\text{-Lipschitz} \iff |f(x) - f(y)| \le L\|x-y\| \quad \forall x, y$

$\Uparrow$

$f(x) - L\|x-y\| \le f(y) \le f(x) + L\|x-y\|$



Wasserstein GAN

$$W(p_{data}(x) \,||\, p_{gen}(x)) \to \min_{p_{gen}}$$

$$\Big\Vert$$

$$\max_{f \in 1\text{-Lipschitz}} \mathbb{E}_{x \sim p_{data}(x)} f(x) - \mathbb{E}_{z \sim N(0,I)} f(g(z)) \to \min_{g}$$

$f(x, \theta)$ – NN, critic

$h(f(x))$, $f(x) \in 1\text{-Lipschitz}$, $h(y) \in 1\text{-Lipschitz} \Rightarrow$

$\Rightarrow h(f(x)) \in 1\text{-Lipschitz}$

$$|h(f(x_1)) - h(f(x_2))| \overset{h \in 1\text{-L}}{\leq} |f(x_1) - f(x_2)| \overset{f \in 1\text{-L}}{\leq} \|x_1 - x_2\|$$

$f(x) = Wx$ $\quad \|Wx_1 - Wx_2\|_2 = \|W(x_1 - x_2)\|_2 \leq$

$$\leq \|W\|_2 \cdot \|x_1 - x_2\|_2$$

$$\|W\|_2 = \max_{x : \|x\| \neq 0} \frac{\|Wx\|_2}{\|x\|_2} = \sigma_{max}(W) = \sqrt{\lambda_{max}(W^T W)}$$

Spectral Normalization

$$W \Rightarrow W_{SN} = \frac{W}{\sigma_{max}(W)} \Rightarrow f \in 1\text{-Lipschitz}$$

$$A = A^T \in \mathbb{R}^{n \times n}, \quad \lambda_{max}(A) - ?$$

Power Iteration: $x \sim N$ $\qquad \lambda_{max} = q_{max}^T A q_{max}$

$$q = \frac{A^k x}{\|A^k x\|} \xrightarrow{k \to +\infty} q_{max}, \text{ where: } A q_{max} = \lambda_{max} q_{max}$$

Eigenvalue decomposition

$$A = Q \Lambda Q^T, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$$

$$Q = [q_1 | \dots | q_n] \in \mathbb{R}^{n \times n}$$

$$Q^T Q = I$$

$$A^2 = Q \Lambda Q^T Q \Lambda Q^T = Q \Lambda^2 Q^T$$

$$A^k = Q \Lambda^k Q^T$$

$$y = A^k x = Q \Lambda^k Q^T x = Q \Lambda^k Q^T \left(\sum_{i=1}^{n} x_i q_i \right) = Q \Lambda^k \left(\sum_i x_i \underbrace{Q^T q_i}_{e_i} \right) =$$

$$= \sum_i x_i Q \Lambda^k e_i = \sum_i x_i q_i (\lambda_i)^k$$

$$\lambda_1 > \lambda_2 > \dots > \lambda_n$$

$$y = \lambda_1^k \left(x_1 q_1 + \sum_{i=2}^{n} x_i q_i \left(\frac{\lambda_i}{\lambda_1} \right)^k \right)$$

$$\frac{A_1^k (x_1 q_1 + \dots)}{\| A_1^k (x_1 q_1 + \dots) \|_2} \xrightarrow{k \to \infty} q_1$$

$v \sim \mathcal{N}$

$$u = \frac{Wv}{\|Wv\|_2}$$

$$v = \frac{W^T u}{\|W^T u\|_2}$$

$$\sigma_{max}(W) \approx u^T W v$$

u, v — buffers in each linear layer

AdaA

T_p – augmentation with power p

$$L_d = \frac{1}{2}\mathbb{E}_{x\sim p_{data}(x)}\left(d(T_p(x)) - 1\right)^2 + \frac{1}{2}\mathbb{E}_z\left(d(T_p(g(z))) - 0\right)^2 \to \min_d$$

$$r_t = \mathbb{E}_{x \sim p_{data}} \operatorname{sign}\left(D_p(x)\right)$$

$$r_{target} = 0.6$$

$$error = r_t - r_{target}$$

$$p = \max\left(0, p + \eta \operatorname{sign}(error) \frac{N_{batch}}{N_{total}}\right)$$